Lab: Data Orchestration using AWS Step Functions (State Machine for Batch Pipeline)

Summary of Lab:

In this lab, we orchestrate the complete batch pipeline using **AWS Step Functions**.

We create a **State Machine** that:

1. Runs MySQL extraction Glue jobs
2. Loads data into Redshift Raw layer
3. Loads data into Redshift Process layer

The workflow executes all Glue jobs in sequence, forming an end-to-end batch data pipeline.

Architecture Overview:

The state machine executes the following flow:

1. MySQL Extraction – Apartments
2. Redshift Raw Ingestion – Apartments
3. MySQL Extraction – Apartment Viewings
4. Redshift Raw Ingestion – Apartment Viewings
5. MySQL Extraction – Apartment Attributes
6. Redshift Raw Ingestion – Apartment Attributes
7. Redshift Process Layer Ingestion

Each step is a **Task state** that triggers a Glue job.

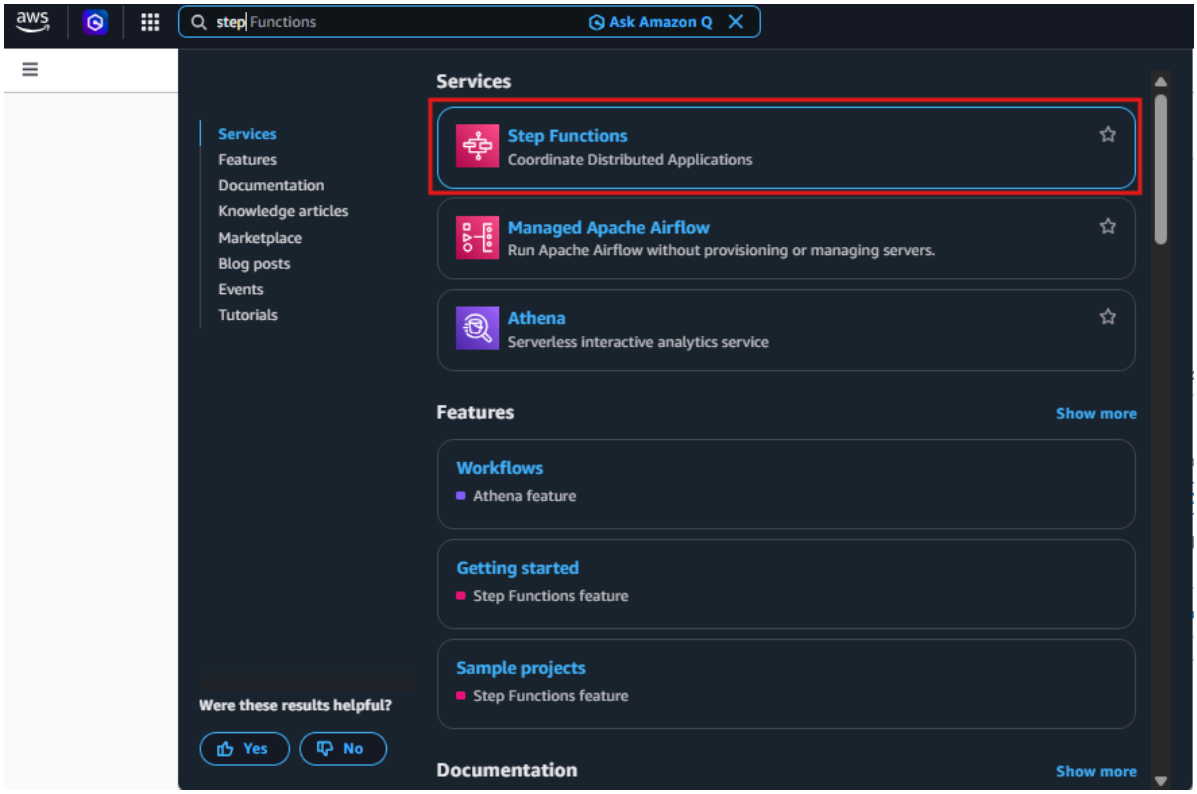
Prerequisites:

- Glue Jobs already created:

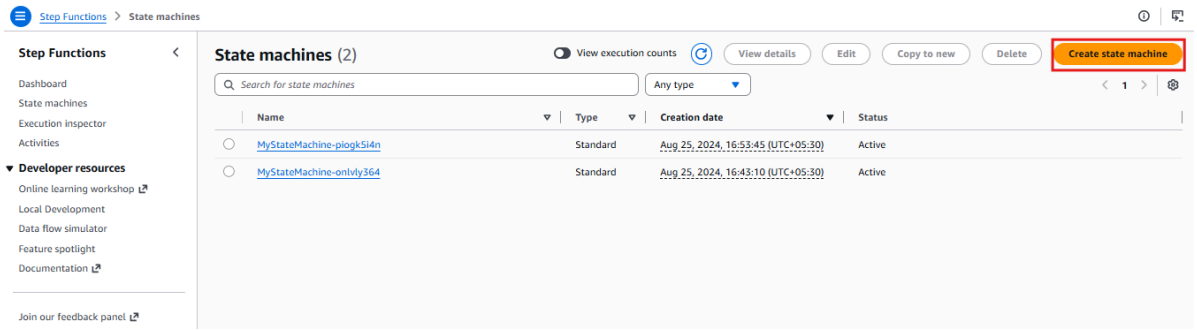
    - `MySQL extraction job`
    - `Redshift raw ingestion`
    - `Redshift process ingestion`
- IAM role for Step Functions (to be created in this lab)
- Step function JSON definition file available (`step_functions.json`)
- All Glue jobs tested and working individually

Step 1: Create the State Machine

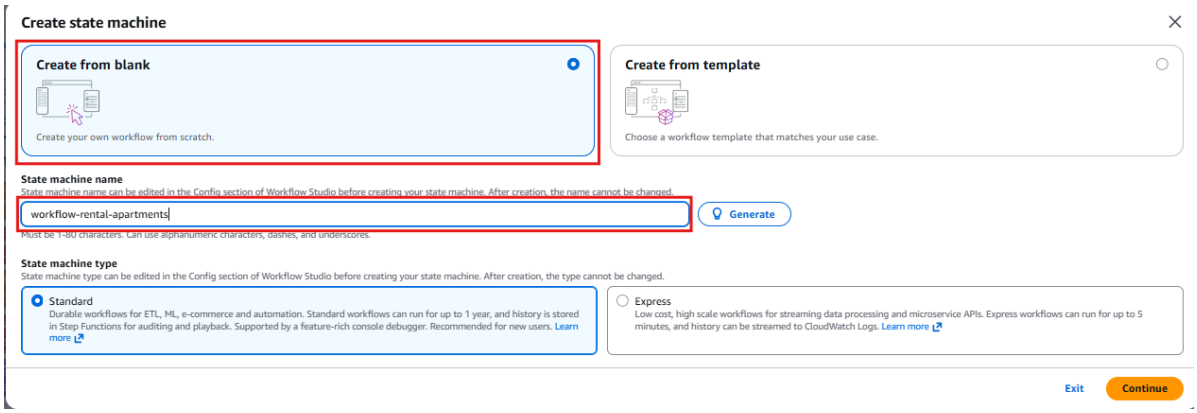
1. Go to AWS Console —→ Search for **Step Functions**



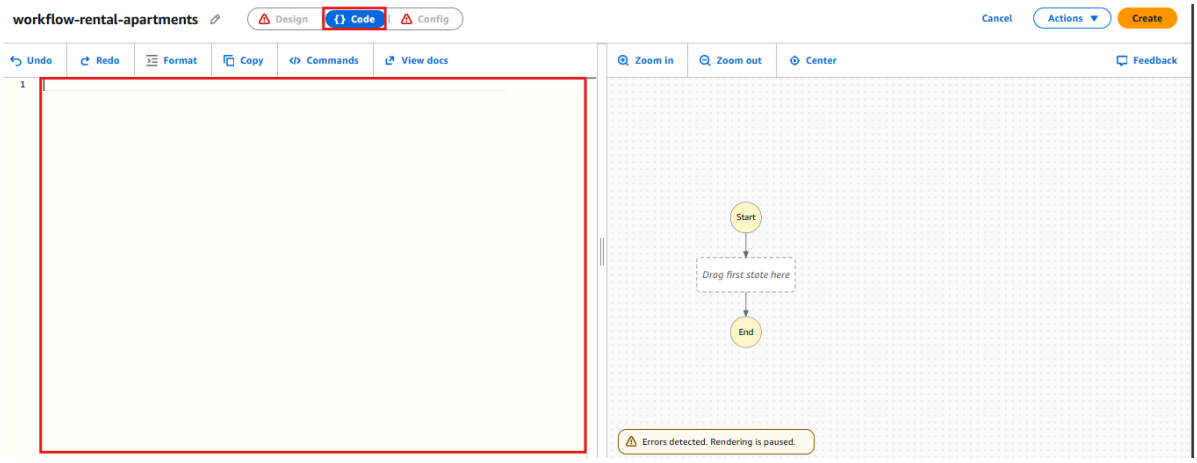
2. Click **Create state machine**



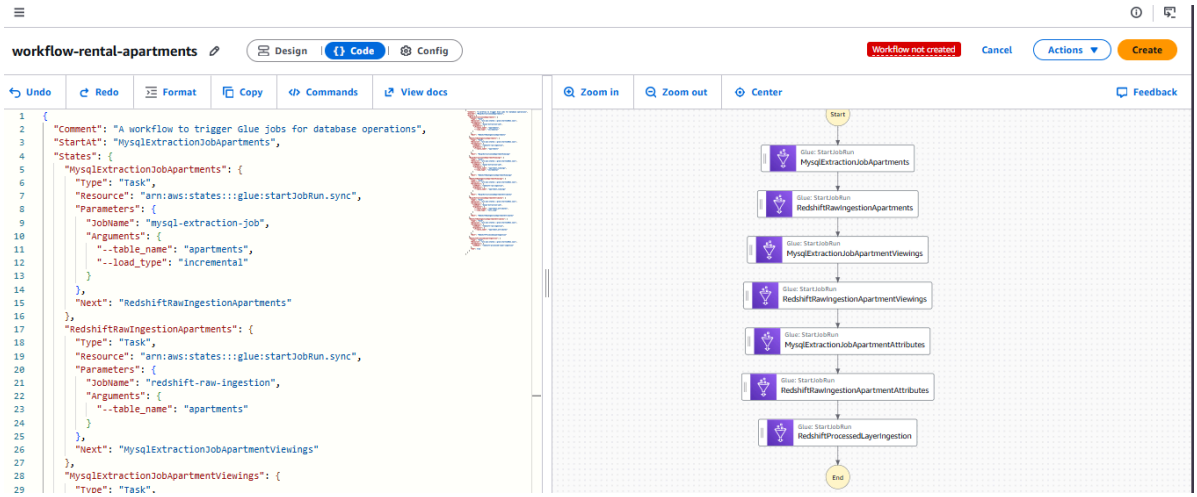
3. Select **Blank —→ Give a name**



4. Go to the **Code tab**

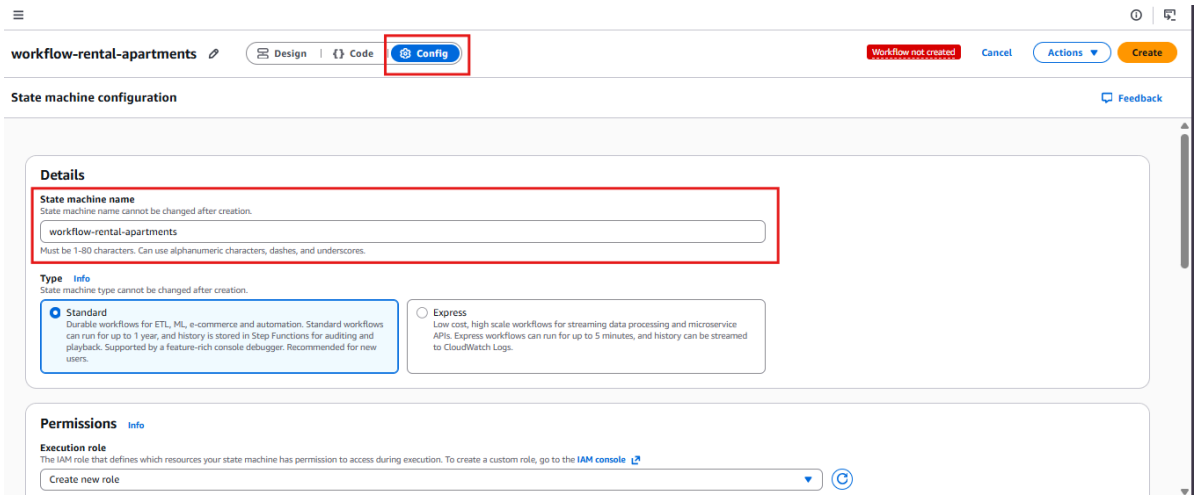


5. Paste the entire JSON definition

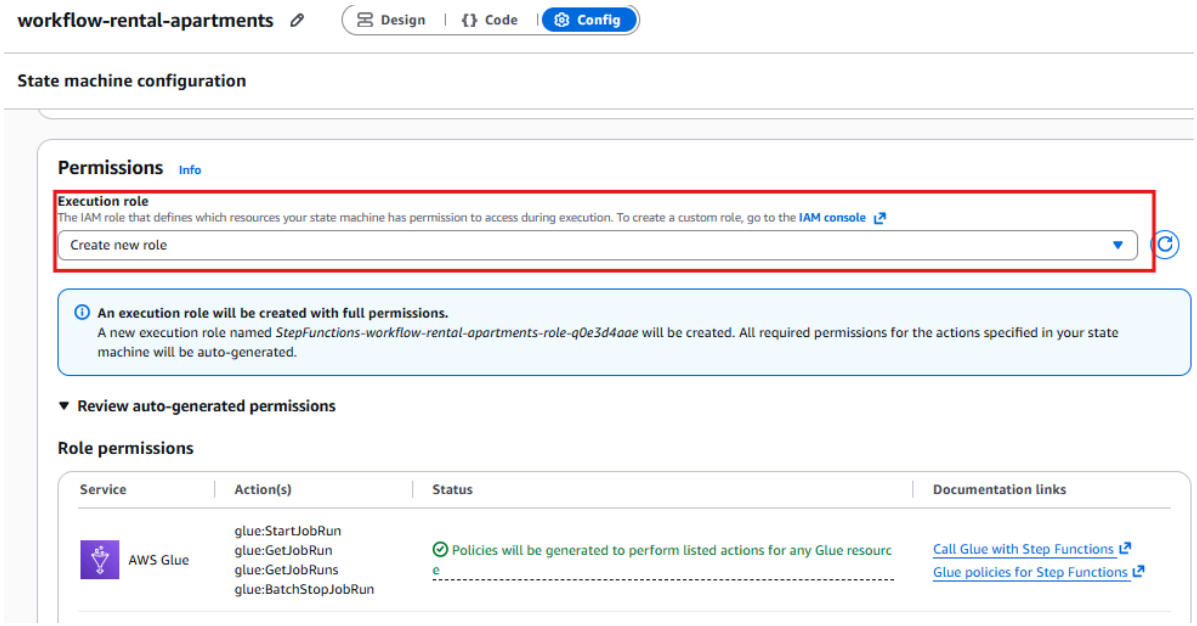


Step 2: Configure State Machine

- Name: `Rental_Apartments_Workflow`

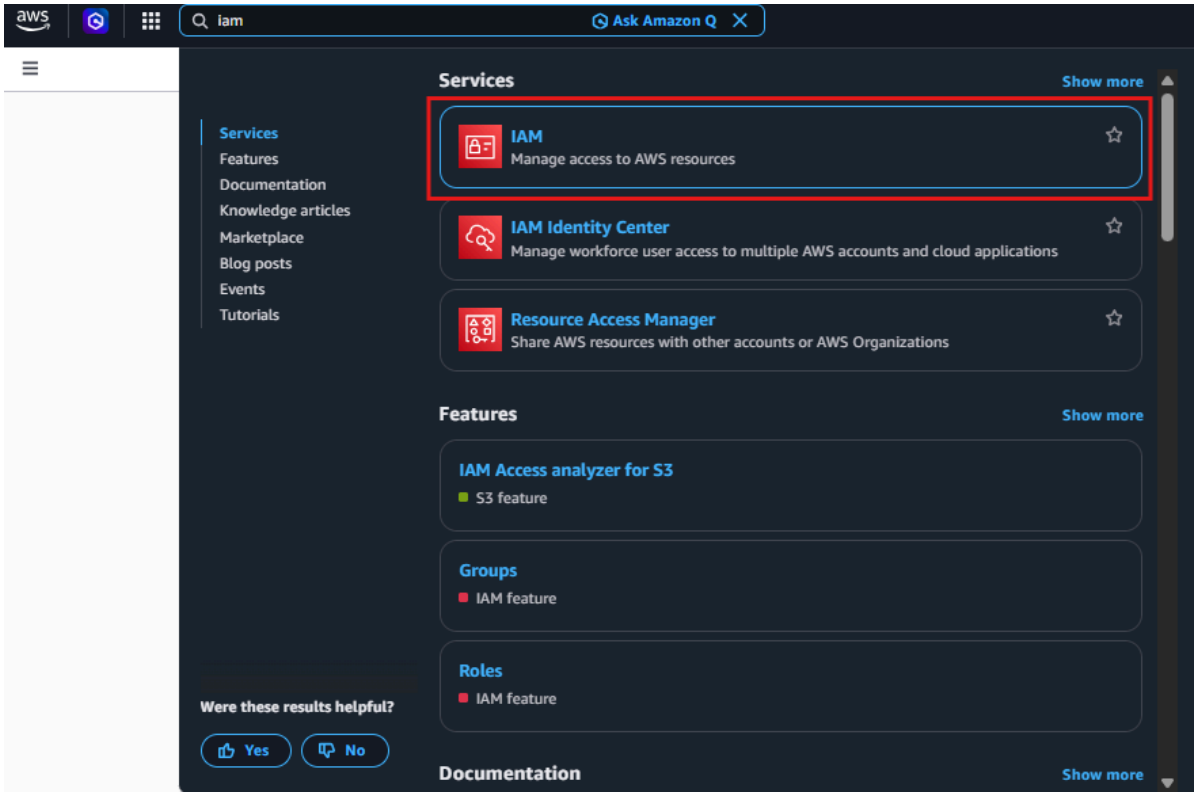


- Now we need to create and attach a custom IAM role.

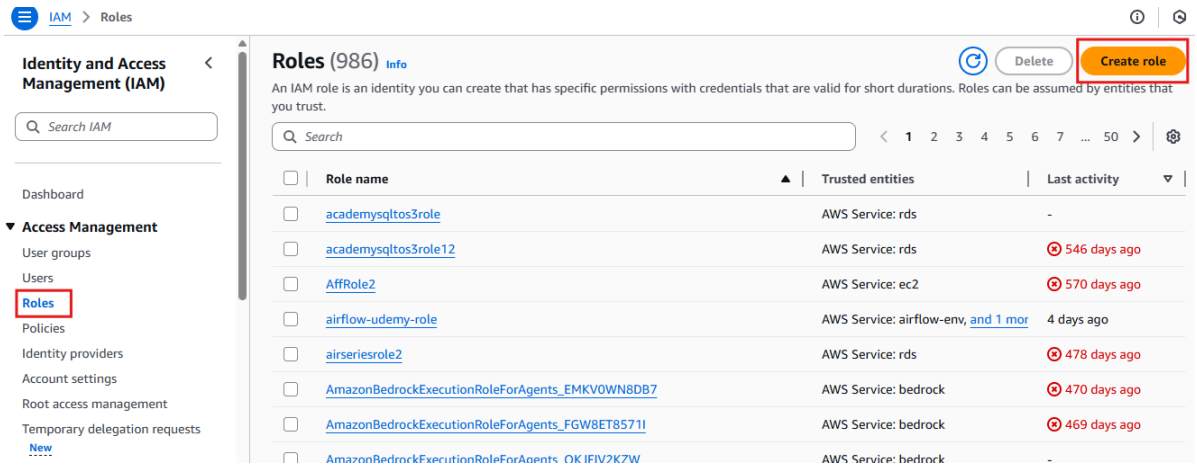


Step 3: Create IAM Role for Step Functions

1. Go to IAM → Roles

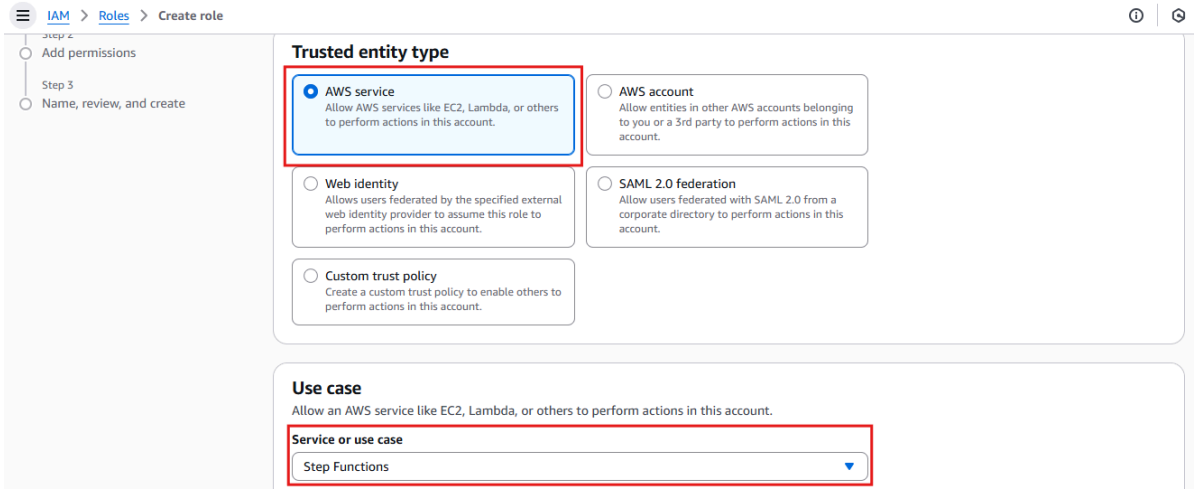


2. Click **Create Role**



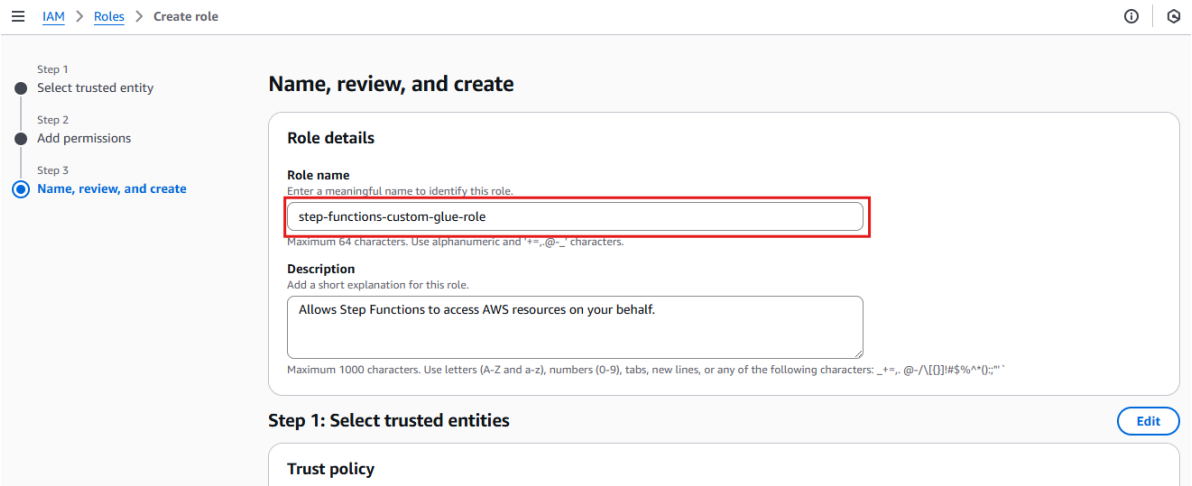
3. Select:

   - Trusted entity type: **AWS service**
   - Use case: **Step Functions**

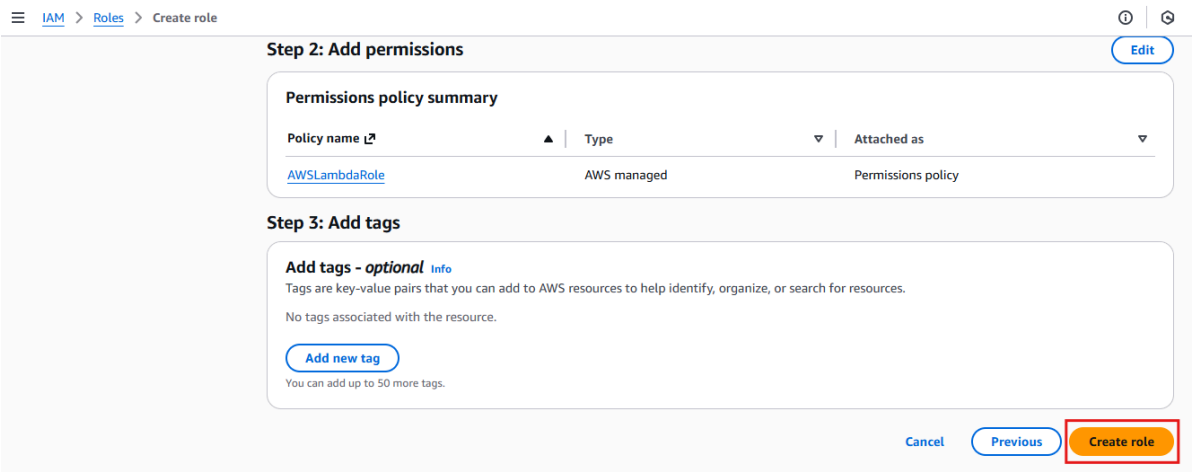


4. Name the role:

```
Custom_StepFunctions_Glue_Role
```

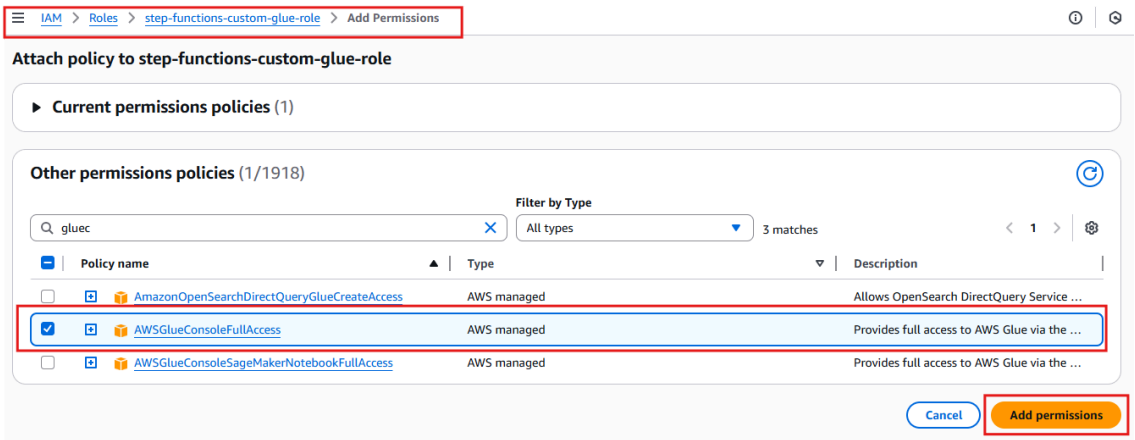


5. Click **Create Role**

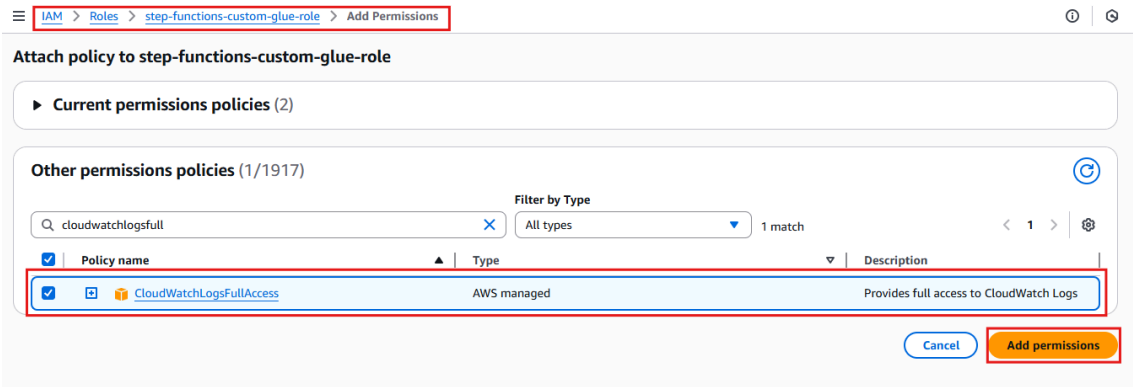


6. Attach Policies:

- `AWSGlueConsoleFullAccess`

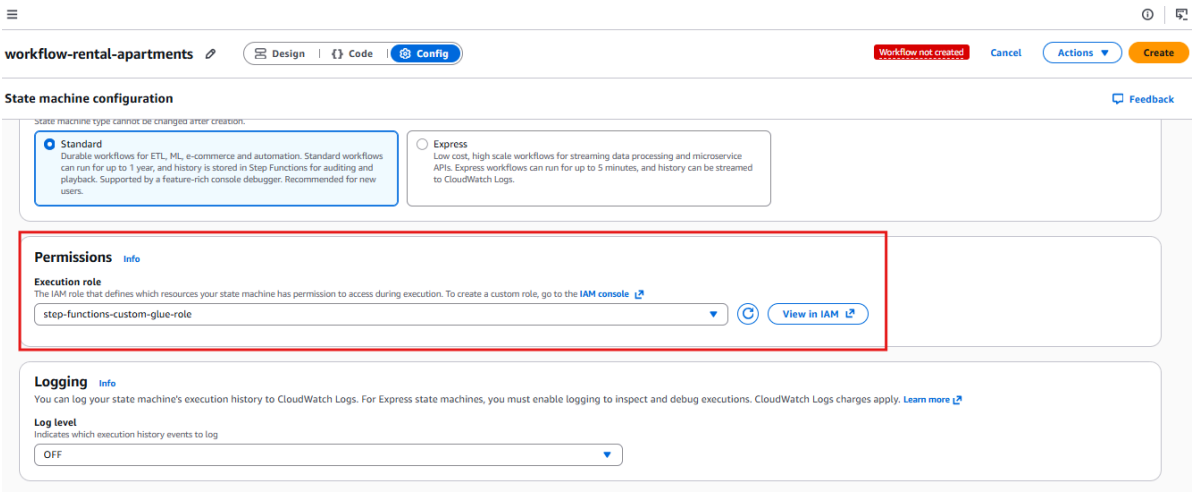


- `CloudWatchLogsFullAccess`

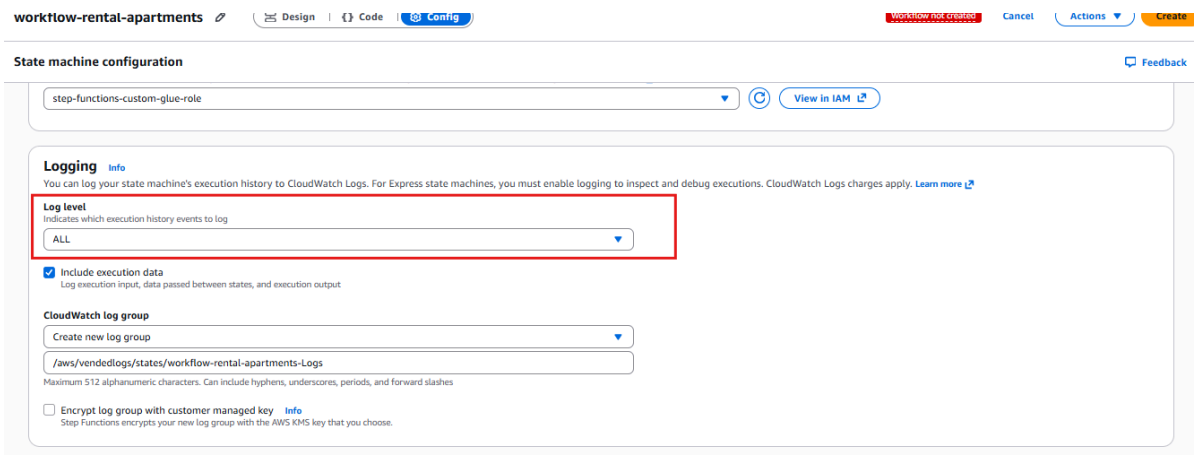


Step 4: Attach Role to State Machine

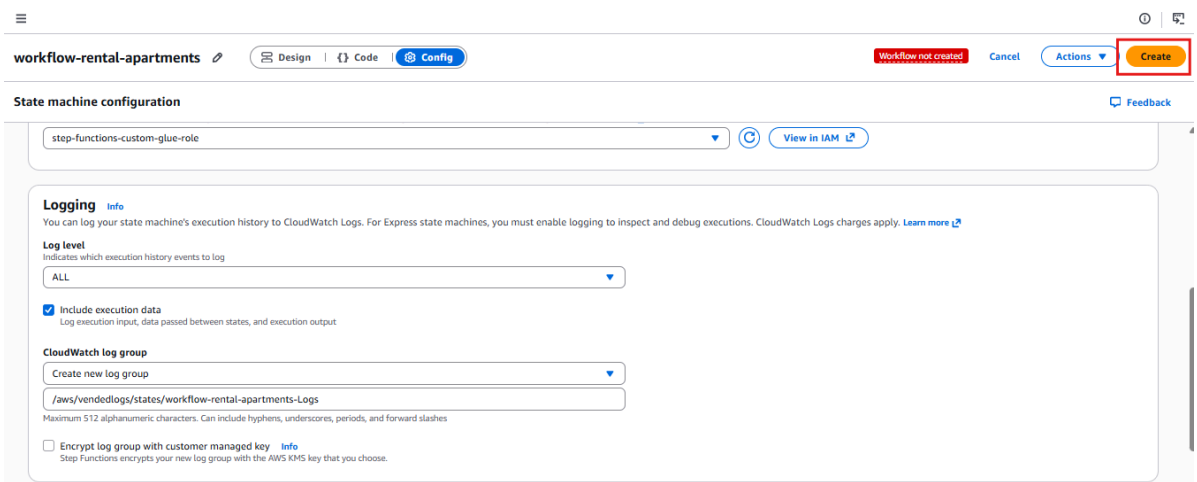
- Go back to Step Functions
- Under Configuration → Select the newly created role



- Log level: All

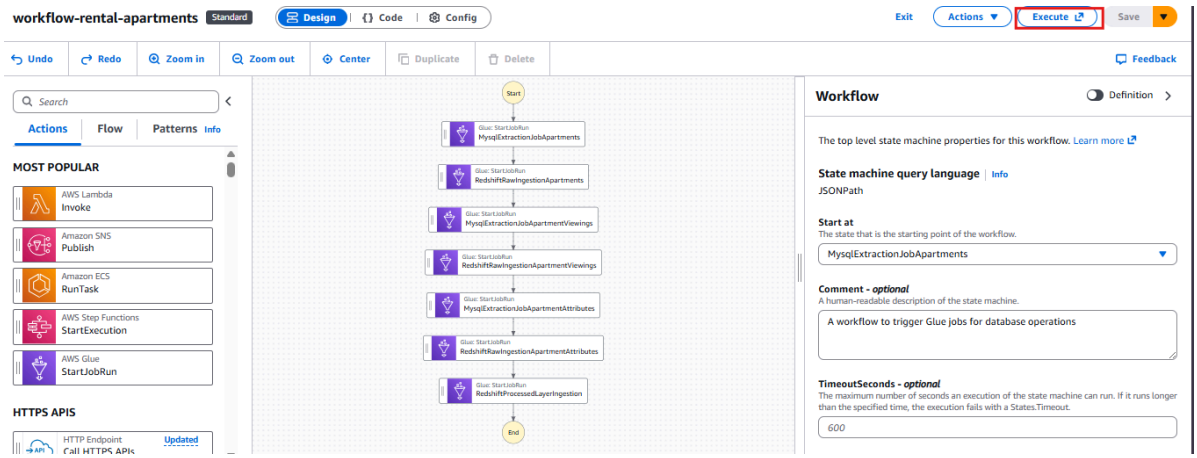


- Click **Create**

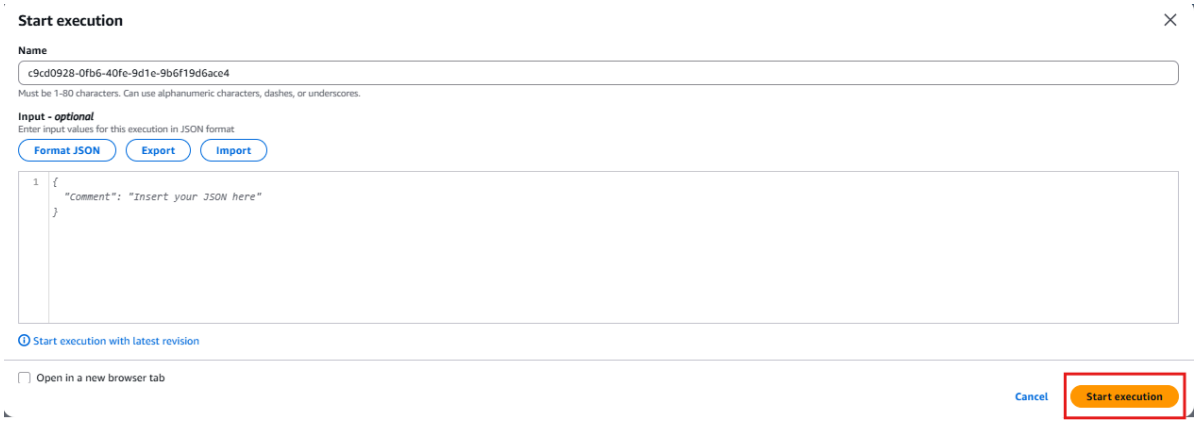


Step 5: Execute the Workflow

1. Click **Execute**



2. Leave input empty
3. Click **Start Execution**



Step 6: Monitor Execution

You can monitor in two ways:

A. Step Functions Graph View

- Observe each state turning green as it completes
- See transitions from extraction → raw ingestion → process ingestion

B. Glue Console

- Navigate to Glue → ETL Jobs
- Check **Runs** for:
  - MySQL extraction job
  - Redshift raw ingestion job
  - Redshift process ingestion job

Each job will be triggered automatically by the state machine.

Expected Execution Flow:

```
MySQL Extraction (apartments)
          ↓
```

```
Redshift Raw Ingestion (apartments)
        ↓
MySQL Extraction (apartment viewings)
        ↓
Redshift Raw Ingestion (apartment viewings)
        ↓
MySQL Extraction (apartment attributes)
        ↓
Redshift Raw Ingestion (apartment attributes)
        ↓
Redshift Process Layer Ingestion
```

Verification / Outcome:

- All Glue jobs triggered sequentially
- All states turn green in Step Functions
- Data refreshed in:
  - `raw_zone` tables
  - `process_zone` tables
- End-to-end batch pipeline executes successfully
- Total execution time: ~10 minutes (approx.)

Pipeline Status After This Lab

You now have:

1. Extraction layer (MySQL → S3)
2. Raw ingestion layer (S3 → Redshift Raw)
3. Process layer ingestion (Raw → Star Schema)
4. Orchestration using Step Functions